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B.Tech. Degree I & II Semester Supplementary Examination November 2023

**CE/EE/ME/SE 19-200-0102 A / CS/EC/IT 19-200-0202 B ENGINEERING CHEMISTRY
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Interpret the basic principles and concepts of quantum mechanics.
 CO2: Account for how spectroscopic methods can be used to determine molecular structures, with focus on the identification of characteristic groups in polyatomic molecules.
 CO3: Apply the laws of thermodynamics to engineering systems.
 CO4: Explain the chemistry of a few important engineering materials and their industrial applications.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

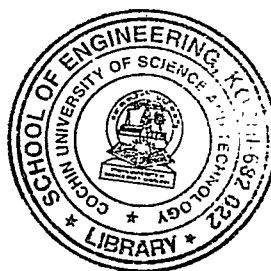
	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	Write down the hydrogen atom wavefunctions and explain the terms.	3	L1	1	3
(b)	Explain Hamiltonian operator.	3	L2	1	2
(c)	Explain Fluorescence with Jablonsky diagram.	3	L2	2	3
(d)	In UV-vis spectra of $\text{CH}_3\text{-CO-CH}_3$ and $\text{CH}_2\text{=CH-CO-CH}_3$, which one will exhibit higher value of λ_{max} . Explain.	3	L2	2	2
(e)	Derive Kirchoff's relation.	3	L1	3	3
(f)	Explain Reduced Phase rule.	3	L1	3	3
(g)	Write a note on flash point and fire point.	3	L1	4	2
(h)	Explain the mechanism of polymerisation initiated by free radical.	3	L1	4	2

PART B

(4 × 12 = 48)

II. (a)	Imagine that an electron is continuously moving in x direction only in a box of ∞ height and length a. Develop an acceptable wavefunction and determine energy states of electron.	8	L2	1	3
(b)	Explain the concept of aromaticity with example.	4	L2	1	3
OR					
III. (a)	Draw the molecular orbital diagram of Nitrogen molecule. Find out the Bond order.	10	L2	1	3
(b)	Explain the aromatic behaviour of Naphthalene.	2	L2	1	3

(P.T.O.)



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		Marks	BL	CO	PO
IV.	(a) Explain the concept of spin-spin splitting with the example of Ethanol.	6	L2	2	3
	(b) Explain the use of rotational spectra in determining the bond length of a diatomic molecule from its pure rotational spectrum.	4	L3	2	2
	(c) Explain the low resolution and high resolution NMR spectra of the following: (i) CH_3CHO (ii) $\text{CH}_3\text{CH}_2\text{SH}$.	2	L2	2	2
OR					
V.	(a) Which type of nuclei are NMR active? Under what conditions such nuclei will be in resonance with radiation absorbed.	8	L3	2	3
	(b) Explain the theory of XPS.	4	L2	2	3
VI.	(a) Explain the Lead-silver system with a neat labelled phase diagram.	8	L2	3	2
	(b) The free energy change (ΔG) accompanying a given process is -86 KJ at 298 K and -84 KJ at 308 K. Calculate the enthalpy change for the process at 303 K.	4	L3	3	1
OR					
VII.	(a) Derive Gibbs-Helmholtz equation. Explain mathematically how free energy is related to pressure and temperature.	8	L2	3	3
	(b) What is the equilibrium constant K_{eq} for the following reaction at 400 K. $2\text{NOCl} \rightleftharpoons 2\text{NO(g)} + \text{Cl}_2\text{(g)}$ given that $\Delta H^\circ = 77.2 \text{ KJ mol}^{-1}$; and $\Delta S^\circ = 122 \text{ JK}^{-1} \text{ mol}^{-1}$.	4	L3	3	3
VIII.	Write a note on moulding techniques of plastics.	12	L1	4	3
OR					
IX.	(a) What are refractions? How are they classified?	3	L1	4	3
	(b) Write a note on synthetic polymers.	6	L1	4	3
	(c) Write a note on thermoplastics and thermosetting plastics.	3	L1	4	3

Bloom's Taxonomy Levels

L1 – 34.6%, L2 – 50%, L3 – 15.4%.
